

Bubble-Wrap and Box a Fragile Item SOP

This SOP covers packing a single fragile item: wrapping it in two layers of bubble wrap, taping both open ends of the wrap, placing it in a box on a foam pad with a second pad on top, applying a fragile sticker, sealing the box and applying the shipping label, using a two-arm robot system. Both arms are used throughout: the grippers cooperate to wrap, tape, box, seal and label, with the left and right grippers taking the specific roles called out in each step. The task runs from an unwrapped item at the back-left of the table to a sealed, stickered and labelled carton at the back-right.

What stays constant across all sessions:

- **Working position:** the item is always moved to the centre of the table, onto the bubble wrap sheet, before any wrapping begins.
- **Wrap target:** every item must be wrapped in **two full layers of bubble wrap, bubbles facing the item**, with no part of the item showing through the wrap, before it goes into the box.
- **Pack order:** wrap, tape the ends, pad base, item, pad top, fragile sticker, seal, label. The order never changes and no step is skipped.
- **Tape rule:** the two ends of the wrap are taped before the bundle goes in the box; the box seam is taped only after the top pad is in.
- **Sticker and label placement:** the **fragile sticker goes on the front face** of the box and the **shipping label goes on the top face**, and neither ever overlaps the other or the seal seam.
- **One at a time:** exactly one item, one wrap sheet, one box, two pads, one fragile sticker and one shipping label are handled per episode.

Setup

Go through both checklists before starting the episode.

Hardware checklist

- ☐ Cameras are on and recording
- ☐ Env camera frame includes the front and back edges of the table and is centered on the table's midpoint
- ☐ Both arms are at the home position with grippers open
- ☐ Table surface is clear of any objects other than the bubble wrap sheet, the box, the pad stack, the tape dispenser, the sticker sheet, the label sheet, the waste tray and the item(s)

Materials checklist

- ☐ Item(s) to be packed are at the **back-left** of the table, upright and not touching each other

- ☐ One cut bubble wrap sheet is laid flat in the working area at the **center, bubble side up**, long edge running left to right
- ☐ The open box sits at the **front-center** with all four flaps folded outward and clear of the opening
- ☐ Foam pads are in a single stack at the **front-left** (at least two pads available)
- ☐ The tape dispenser is at the **right edge** with a free tape end exposed and reachable
- ☐ The fragile sticker sheet is at the **front-right** with the next sticker peeled and exposed
- ☐ The shipping label sheet is at the **front-right**, behind the sticker sheet, with the next label peeled and exposed
- ☐ The waste tray is at the **right edge** and is empty or only part full
- ☐ Back-right of the table is clear (output location for the finished carton)

Workspace layout

- **Back-left**: unwrapped fragile item(s) (input)
- **Center** : bubble wrap sheet - working area (wrap and tape)
- **Front-center**: open box (pad, item, pad, seal)
- **Front-left**: foam pad stack
- **Front-right**: fragile sticker sheet and shipping label sheet
- **Right edge**: tape dispenser and waste tray (wrap offcuts and backing paper only)
- **Back-right**: finished sealed carton (output)

Vocabulary

These are the terms used in this SOP. Operators and annotators must use this language consistently. One term per concept, used throughout.

Item anatomy

- **Item**: a single fragile object to be wrapped and boxed.
- **Body**: the main mass of the item, between its two ends.
- **End**: one of the two faces of the item that the wrap does not cover on the first pass.
- **Protrusion**: any handle, spout, stem or corner that sticks out from the body.
- **Contact face**: the face of the item that rests on the wrap sheet at the start.

Packaging anatomy

- **Bubble wrap sheet**: the pre-cut sheet of bubble wrap one item is wrapped in.
- **Bubble side**: the face of the sheet carrying the raised bubbles.

- **Flat side:** the smooth back face of the sheet, opposite the bubble side.
- **Layer:** one full turn of the sheet around the body of the item.
- **Wrap seam:** the line where the trailing edge of the sheet lands on the second layer.
- **Wrap end:** one of the two open ends of the wrap tube, at either end of the item.
- **Bundle:** the item once it is wrapped in two layers and both ends are taped.
- **Box:** the open carton the bundle goes into.
- **Box flap:** one of the four top panels of the box that fold over to close it.
- **Seal seam:** the line where the two long box flaps meet when the box is closed.
- **Foam pad:** one cushioning pad; one goes under the bundle and one on top of it.
- **Pad base:** the foam pad lying flat on the bottom of the box, under the bundle.
- **Pad top:** the foam pad lying flat on top of the bundle, under the flaps.
- **Tape strip:** one length of tape drawn from the dispenser and pressed down.
- **Fragile sticker:** the adhesive sticker marking the carton as fragile.
- **Shipping label:** the adhesive label carrying the destination details.
- **Carton:** one sealed, stickered and labelled box containing one wrapped item.
- **Box front face:** the vertical face of the box nearest the operator.
- **Box top face:** the upward face of the box once the flaps are closed.

Workspace zones

- **Back-left edge:** input zone holding the unwrapped item(s) at the back-left of the table.
- **Working area:** the center of the table where the bubble wrap sheet lies and where wrapping and taping happen.
- **Box station:** the open box at the front-center of the table where padding, loading and sealing happen.
- **Pad stack:** the foam pad stack at the front-left of the table.
- **Label station:** the sticker sheet and label sheet at the front-right of the table.
- **Tape station:** the tape dispenser at the right edge of the table.
- **Waste tray:** the tray at the right edge that takes wrap offcuts and backing paper and nothing else.
- **Back-right edge:** output zone where the finished carton is placed at the back-right of the table.
- **Home position:** the default resting pose for each arm: gripper open and clear of the table.

Actions

- **Pick:** lift one item clear of the table and carry it to the working area.
- **Square:** adjust the item so it lies centered on the wrap sheet with its long axis running left to right.
- **Roll:** turn the item over the sheet so the sheet takes up one full layer around the body.
- **Tuck:** fold the loose wrap at one wrap end in against the end of the item.
- **Tape:** draw one tape strip from the dispenser and press it down over its full length.
- **Load:** lower the bundle into the box onto the pad base without dropping it.
- **Seal:** press a tape strip along the seal seam so the box stays closed.
- **Apply:** press the fragile sticker or the shipping label flat onto the named box face.

Steps

Step 1: Move an item onto the bubble wrap sheet

Goal: one item rests on the bubble wrap sheet in the working area, ready to be squared.

- With the left gripper, grasp one item from the back-left by its body, not by a protrusion.
- Lift it clear of the table and carry it to the center; do not drag or slide it.
- Lay it down on the bubble wrap sheet and release.

Step 2: Square the item on the sheet

Goal: the item lies centered on the sheet, long axis left to right, with enough sheet on the near side to reach two full layers.

2.1 Check the sheet orientation

- If the bubble side is up → continue to 2.2.
- If the flat side is up → lift the item clear, grasp the two far corners of the sheet with both grippers, turn the sheet over so the bubble side is up, then lay the item back down.

2.2 Center the item on the sheet

- With both grippers, adjust the item so its long axis runs left to right and it sits on the far half of the sheet, 2-3 cm inside the far edge.
- Confirm the near side of the sheet is long enough to pass around the body twice. If it is not, move the item farther toward the far edge.

2.3 Check the wrap ends

- Confirm the sheet extends past each end of the item by at least half the item's thickness, so both wrap ends can be tucked.
- If one end is short, slide the item along the sheet until the overhang is even at both ends.

Step 3: Roll the first layer

Goal: one full layer of bubble wrap is around the body of the item, bubbles against the item.

- With the left gripper, grasp the near edge of the sheet at the left of the item, and with the right gripper, grasp the same edge at the right of the item.
- Lift the near edge and bring it over the item onto the far side in one smooth motion, so the sheet takes up one full layer around the body.
- Press the edge down against the far side of the item and release.
- Confirm the bubbles face the item and the layer is snug with no air gap along the body.

Step 4: Roll the second layer

Goal: a second full layer is around the item, with the trailing edge pressed down as the wrap seam.

- With both grippers, grasp the wrapped item near each end and roll it forward over the remaining sheet until the sheet is used up.
- Press the trailing edge down onto the wrap to form the wrap seam.
- Confirm two full layers cover the body and no part of the item shows through the wrap.

Warning: One layer is not enough and three is not the target. If the wrap comes up short of two full layers, unroll it and start again with the item farther toward the far edge of the sheet.

Step 5: Tape the wrap seam and both wrap ends

Goal: the wrap seam and both wrap ends are taped, so the wrap cannot open on its own.

5.1 Tape the wrap seam

- With the left gripper, hold the wrapped item down in the working area.
- With the right gripper, draw one tape strip from the dispenser at the right edge and press it down along the wrap seam over its full length.

5.2 Tuck and tape the left wrap end

- With the left gripper, fold the loose wrap at the left wrap end in against the end of the item, working around until the end is closed.
- With the right gripper, draw one tape strip and press it across the tucked end so it holds the tuck flat.

5.3 Tuck and tape the right wrap end

- With the right gripper, fold the loose wrap at the right wrap end in against the end of the item, working around until the end is closed.
- With the left gripper, draw one tape strip and press it across the tucked end so it holds the tuck flat.
- Release both grippers.

Warning: Both wrap ends must be taped before the bundle goes in the box. A taped seam with open ends does not count as wrapped.

Step 6: Place the pad base in the box

Goal: one foam pad lies flat on the bottom of the box, covering it corner to corner.

- With the right gripper, grasp the top pad of the front-left stack.
- Lift exactly one pad. If two or more lift together, set them back down and re-grip.
- Lower it into the box and lay it flat on the bottom, then release.
- With either gripper, press the pad down at its center so it sits flat with no lifted corner and no gap at the box walls.

Step 7: Load the bundle into the box

Goal: the bundle sits centered on the pad base, below the top edge of the box.

- With the left gripper, grasp one end of the bundle, and with the right gripper, grasp the other end.
- Lift the bundle clear of the table, carry it over the box, and lower it onto the pad base in one continuous motion. Do not drop it in.
- Center the bundle in the box, long axis running left to right, with clearance on all four sides.
- Release both grippers.
- Confirm the top of the bundle sits below the top edge of the box, so the flaps can close over it.

Warning: The bundle must be lowered onto the pad, not dropped. A dropped bundle is a violation even if the item is undamaged.

Step 8: Place the pad top over the bundle

Goal: the second foam pad lies flat on top of the bundle, inside the box and below the flap line.

- With the right gripper, grasp the next pad from the front-left stack and lift exactly one.
- Lower it into the box onto the bundle and lay it flat, then release.
- Press the pad down at its center so it lies flat and no corner sticks up past the top edge of the box.

Step 9: Apply the fragile sticker

Goal: the fragile sticker is stuck flat on the front face of the box, before the box is sealed.

- With the right gripper, take the next exposed fragile sticker from the sheet at the front-right.
- Land it on the box front face, centered side to side, in the upper third, reading the right way up from the operator's side (within 2 cm and 10°).
- Press the sticker down over its full area and release.

- If a corner lifts, press it down again. Do not peel the sticker off and re-apply it, and do not add a second sticker.

Step 10: Close the flaps and seal the box

Goal: all four flaps are closed and the seal seam is taped over its full length.

10.1 Close the flaps

- With the left and right grippers, fold the two short flaps in first, one with each gripper.
- Fold the near long flap in with the right gripper, then the far long flap in with the left gripper, so the two long flaps meet along the seal seam.
- Confirm no pad, wrap or tape is trapped between the flaps.

10.2 Tape the seal seam

- With the left gripper, hold the far long flap down at the seam.
- With the right gripper, draw one tape strip from the dispenser and press it down along the seal seam from one end of the box to the other.
- Release both grippers.

Warning: The box is not sealed until the tape runs the full length of the seal seam. Flaps that are closed but untaped count as unsealed.

Step 11: Apply the shipping label

Goal: the shipping label is stuck flat on the top face of the carton, centered, the right way up, and clear of the seal seam and the fragile sticker.

- With the right gripper, take the next exposed shipping label from the sheet at the front-right.
- Land it on the box top face, centered on one side of the seal seam, reading the right way up from the operator's side (within 2 cm and 10°).
- Press the label down over its full area and release.
- Confirm the label does not cross the seal seam and does not overlap the fragile sticker.
- If a corner lifts, press it down again. Do not peel the label off and re-apply it, and do not add a second label.

Step 12: Move the carton to the back-right edge

Goal: the finished carton rests in the designated back-right output zone.

- With the right gripper, grasp the carton along the middle of its right edge.
- Drag it back and to the right into the output location at the back-right of the table.
- Release the gripper.

Step 13: Reset the working area, home the arms and end the episode

Goal: the working area holds the next bubble wrap sheet and both arms are home, ready for the next item.

- With the left gripper, lay the next cut bubble wrap sheet flat in the center of the working area, bubble side up, long edge running left to right.
- With the left gripper, move the next open box to the front-center with all four flaps folded outward.
- Move both arms back to the home position with grippers open.
- End data collection.

SOP violations

These are the things the operator can do that break (violate) the SOP. This list feeds the violation set and is what reviewers look for using the side-by-side review tool.

How to record a violation in review

For each violation you spot in a recorded episode, record:

- The **start timestamp** of the violation in the video.
- The **violation name** from the list below.
- The **SOP rule broken** (the step number from the list below).

The **visible cue** is what you actually see in the video. The **coaching note** is for retraining the operator after the review; it is not what the annotator labels.

Episode handling

Any SOP violation is flagged (or tagged) with the timestamp and violation name. Rather than being discarded, the episode is retained in the training data and tagged with the violation. No flagged episode is deleted. This matches the goal of capturing realistic operator variance in training.

Violations

Violation: Wrong pickup.

- **Visible cue:** the item is started from the center or the right instead of the back-left input zone, it is placed outside the bubble wrap sheet, it is grasped by a protrusion instead of the body, or the item is dragged or slid across the table instead of being picked up (lifted clear of the surface) and carried.
- **SOP rule broken:** Step 1 (pick from the back-left by the body and lay it on the wrap sheet). Pick means lift and carry, not drag.
- **Coaching note:** always start each item from the back-left and bring it to the sheet before squaring. Grasp the body, lift the item clear of the table and carry it. Do not drag or slide it.

Violation: Picked more than one item at once.

- **Visible cue:** two or more items are lifted/moved from the input zone in a single pick.
- **SOP rule broken:** Step 1 (move one item).
- **Coaching note:** pick exactly one item per episode.

Violation: Wrapping while not squared (squaring skipped).

- **Visible cue:** the operator starts to roll the wrap while the item is off the sheet, with its long axis front-to-back, or without enough sheet on the near side for two layers, skipping the checks in Step 2.
- **SOP rule broken:** Step 2 (centered on the sheet, long axis left to right, enough sheet for two full layers, before any roll).
- **Coaching note:** never start rolling straight away. Square the item on the sheet and check the near-side length first.

Violation: Wrapped bubbles-out.

- **Visible cue:** the item is rolled with the flat side of the sheet against it, so the bubbles face outward.
- **SOP rule broken:** Step 2.1 and the wrap target (bubbles facing the item).
- **Coaching note:** check the sheet face before the item goes down. Bubbles always face the item.

Violation: Fewer than two layers of wrap.

- **Visible cue:** the wrap covers the body with only one layer, or the second layer stops short and part of the first layer is still exposed at the end of Step 4.
- **SOP rule broken:** Step 4 (two full layers around the body).
- **Coaching note:** two full turns, every time. If the sheet runs out early, unroll and restart with the item farther toward the far edge.

Violation: Item visible through the wrap.

- **Visible cue:** at the end of Step 4 part of the item is uncovered - a protrusion, a corner, or an end poking out of the wrap.
- **SOP rule broken:** Step 4 (no part of the item shows through the wrap).
- **Coaching note:** walk the wrap around the whole body and check every protrusion before taping.

Violation: Wrap seam not taped.

- **Visible cue:** the trailing edge of the wrap is pressed down but no tape strip is run along the wrap seam, or the tape is pressed at one point only.
- **SOP rule broken:** Step 5.1 (press one tape strip along the wrap seam over its full length).

- **Coaching note:** run the strip end to end along the seam. A spot press does not hold the wrap.

Violation: Wrap end not tucked or not taped.

- **Visible cue:** one or both wrap ends are left open, tucked but untaped, or taped without being tucked in against the end of the item.
- **SOP rule broken:** Steps 5.2 and 5.3 (tuck each wrap end in against the item, then tape it flat).
- **Coaching note:** tuck first, then tape, and do both ends. Open ends let the item slide out of the wrap.

Violation: Bundle boxed before both ends are taped.

- **Visible cue:** the bundle goes into the box while the wrap seam or either wrap end is still untaped.
- **SOP rule broken:** Step 5 (wrap seam and both wrap ends taped before the bundle goes in the box).
- **Coaching note:** finish all three tape strips at the working area. Nothing enters the box half taped.

Violation: Pad base missing or not flat.

- **Visible cue:** the bundle is loaded onto the bare bottom of the box, or the pad base is left folded, crooked, or with a lifted corner or a gap at the box walls.
- **SOP rule broken:** Step 6 (one pad flat on the bottom of the box, corner to corner, pressed down).
- **Coaching note:** the pad goes in first and gets pressed flat. Check for lifted corners before the bundle goes in.

Violation: More than one pad lifted.

- **Visible cue:** two or more pads come off the front-left stack together and the operator continues without setting them back down and re-gripping.
- **SOP rule broken:** Steps 6 and 8 (lift exactly one pad; if two or more lift together, set them back down and re-grip).
- **Coaching note:** separate the top pad from the stack before lifting, and put a double pick straight back down.

Violation: Bundle dropped into the box.

- **Visible cue:** the bundle is released above the box and falls onto the pad base rather than being lowered onto it in a continuous motion.
- **SOP rule broken:** Step 7 (lower the bundle onto the pad base; do not drop it in).

- **Coaching note:** carry the bundle all the way down to the pad before opening the grippers. This is a fragile item.

Violation: Bundle off-center or proud of the box.

- **Visible cue:** at the end of Step 7 the bundle touches a box wall with no clearance, sits on top of the pad edge, or stands above the top edge of the box so the flaps cannot close.
- **SOP rule broken:** Step 7 (centered on the pad base, clearance on all four sides, below the top edge of the box).
- **Coaching note:** center the bundle and check the height before the top pad goes in.

Violation: Pad top missing or not flat.

- **Visible cue:** the box is closed with no pad on top of the bundle, or the top pad is folded, crooked, or sticking up past the top edge of the box.
- **SOP rule broken:** Step 8 (one pad flat on top of the bundle, below the flap line, pressed down).
- **Coaching note:** the second pad is not optional. Lay it flat and press it down before closing the flaps.

Violation: Fragile sticker missing, crooked, or on the wrong face.

- **Visible cue:** the carton leaves Step 9 with no fragile sticker, with the sticker on a face other than the front face, upside down, or more than about 2 cm off center or 10 degrees off square.
- **SOP rule broken:** Step 9 (one fragile sticker, box front face, centered in the upper third, reading the right way up).
- **Coaching note:** aim the sticker before releasing it: front face, upper third, centered, right way up.

Violation: Sticker or label peeled off and re-applied, or doubled.

- **Visible cue:** a placed sticker or label is pulled back off and stuck down again, or a second one is added over or beside the first.
- **SOP rule broken:** Steps 9 and 11 (press a lifted corner back down; do not peel and re-apply, and do not add a second sticker or label).
- **Coaching note:** a lifted corner gets pressed, not removed. One sticker and one label per carton.

Violation: Flaps closed in the wrong order or with something trapped.

- **Visible cue:** the long flaps are folded in before the short flaps, or a pad, the wrap, or a tape end is trapped between the flaps at the end of Step 10.1.
- **SOP rule broken:** Step 10.1 (short flaps first, then near long flap, then far long flap, with nothing trapped between them).

- **Coaching note:** short flaps first, and clear the flap line before the long flaps come over.

Violation: Box closed but not sealed.

- **Visible cue:** the flaps are folded over and released without a tape strip run along the seal seam, or the tape covers only part of the seam.
- **SOP rule broken:** Step 10.2 (press one tape strip along the seal seam from one end of the box to the other).
- **Coaching note:** run the tape the full length of the seam. Closed flaps are not a seal.

Violation: Shipping label missing, crooked, or on the wrong face.

- **Visible cue:** the carton leaves Step 11 with no shipping label, with the label on a face other than the top face, upside down, or more than about 2 cm off center or 10 degrees off square.
- **SOP rule broken:** Step 11 (one label, box top face, centered on one side of the seal seam, reading the right way up).
- **Coaching note:** aim the label before releasing it: top face, one side of the seam, centered, right way up.

Violation: Label crossing the seal seam or overlapping the sticker.

- **Visible cue:** the shipping label is placed across the taped seal seam, or it overlaps the fragile sticker.
- **SOP rule broken:** Step 11 (the label does not cross the seal seam and does not overlap the fragile sticker).
- **Coaching note:** place the label on one side of the seam, clear of the sticker. A label across the seam tears when the box is opened.

Violation: Sequence out of order.

- **Visible cue:** the operator seals the box before the top pad is in, applies the fragile sticker after sealing, tapes the box seam before the wrap ends, or otherwise departs from the pack order.
- **SOP rule broken:** the pack order (wrap, tape the ends, pad base, item, pad top, fragile sticker, seal, label).
- **Coaching note:** follow the pack order every episode. Each step depends on the one before it.

Violation: Waste not placed in the waste tray.

- **Visible cue:** a wrap offcut or a piece of backing paper is dropped on the table, left in the working area, released over the output zone, or another object is put into the waste tray.
- **SOP rule broken:** the waste tray rule (offcuts and backing paper go to the tray at the right edge; nothing else goes in it).

- **Coaching note:** carry the offcut all the way to the tray and release it there. The tray takes offcuts and backing paper only.

Violation: Carton placement off zone.

- **Visible cue:** in Step 12 the carton is moved with the left gripper, the operator doesn't grasp the middle of the right edge while dragging to the output, it is left in the center instead of being dragged to the back-right output zone, or released at the back-right but ending up noticeably off the designated zone, without falling or requiring further correction.
- **SOP rule broken:** Step 12 (grasp the middle of the right edge with the right gripper and drag back and to the right into the back-right output zone).
- **Coaching note:** move the carton to the back-right output zone with the right gripper. Do not leave it in the center and aim the release point more precisely; push the carton deep into the back-right corner.

Violation: Working area not reset.

- **Visible cue:** the episode ends with no bubble wrap sheet in the center, the sheet laid flat side up, or no open box at the front-center with its flaps folded outward.
- **SOP rule broken:** Step 13 (lay the next sheet bubble side up in the center and set the next open box at the front-center).
- **Coaching note:** reset the sheet and the box before homing so the next episode starts from the setup state.

Violation: Wrong arm used for an action.

- **Visible cue:** any step that specifies the left gripper or the right gripper is performed with the opposite gripper.
- **SOP rule broken:** any step that specifies a gripper, including Steps 1, 3, 5, 6, 7, 8, 9, 10, 11, 12 and 13.
- **Coaching note:** operator confusion about left vs right roles. Walk through the SOP step by step with the operator.

Violation: Re-grip on a pick or grasp.

- **Visible cue:** the operator closes on the item, the sheet, a pad, the tape, the sticker or the label, finds the grip off, opens, and re-grips before lifting more than two times.
- **SOP rule broken:** Steps 1/3/5/6/7/8/9/11 (grasp securely, e.g., the body, the sheet edge, the pad, the tape end or the exposed sticker/label).
- **Coaching note:** approach angle off. Practice the from-above approach so the first grip catches the right point.

Violation: Repeated fiddling with placement, wrap or tape.

- **Visible cue:** the operator makes many small adjustments (more than about two) to settle the item, the wrap, a pad, a tape strip, the sticker or the label, rather than settling it in one or two corrections.
- **SOP rule broken:** the square/roll/tape/load/apply sub-steps (settle in one or two small adjustments).
- **Coaching note:** grip or approach is off, forcing repeated correction. Tighten the approach so each roll, each strip and each placement lands close to target.

Violation: Arms not fully at home position at episode end.

- **Visible cue:** in the final frame, both arms are close to the home position but not exactly at it (gripper not fully open, or arm position visibly off home).
- **SOP rule broken:** Step 13 (return to home position with grippers open).
- **Coaching note:** complete the home motion explicitly before ending recording.

Non-violation failures

These are episode failures that are not the operator's fault and do not go in the violation set. They are recorded as system issues, the episode is discarded/deleted, and the episode does not become a coaching point for the operator.

- **Recording stopped or paused mid-episode:** Cause: software or hardware issue with the recording system.
- **Camera dropped frames or lost feed during the episode:** Cause: camera or capture system issue.
- **Hardware fault on the robot arm:** Cause: gripper malfunction, arm position drift, or motor error during the episode.
- **Defective item:** Cause: an item that arrives already chipped, cracked or broken, through no fault of the operator. Replace before the next episode.
- **Defective bubble wrap sheet:** Cause: a sheet cut too short for two layers, or one with the bubbles already crushed flat. Replace before the next episode.
- **Defective box or pads:** Cause: a crushed or split box, flaps that will not hold a fold, or pads torn or too small for the box footprint. Replace before the next episode.
- **Tape dispenser fault:** Cause: tape that will not release, tears off short, or has lost its adhesive. Reload the dispenser before the next episode.
- **Defective sticker or label:** Cause: a sticker or label that will not release from the sheet or has lost its adhesive. Replace the sheet before the next episode.

After each episode: repeat or reset

- **Batch sessions (e.g., 5x):** if fewer than the target number are packed, return to Step 1 with the next item. Once the target number is packed, reset the workspace before the next session.
- Clear the working area and re-run the Setup checklist before the next session.

After the episode: reset the workspace

This part is not recorded. It is just how you reset the table for the next episode.

- With recording off, take the finished carton(s) from the back-right.
- Cut each carton open, take the bundle out, unwrap the item, and stand it at the back-left, so the input zone again matches the initial setup state.
- Discard the used wrap, tape, stickers and labels, and set the used pads aside if they are still flat and undamaged.
- Restock the pad stack at the front-left, expose the next fragile sticker and the next shipping label at the front-right, and confirm the tape dispenser has a free end exposed.
- Lay a fresh cut bubble wrap sheet flat in the center, bubble side up, long edge running left to right.
- Set a fresh open box at the front-center with all four flaps folded outward and clear of the opening.
- Empty the waste tray at the right edge.
- Confirm the table surface is clear of any objects other than the wrap sheet, the box, the pad stack, the tape dispenser, the sticker sheet, the label sheet, the waste tray and the item(s) in their designated zones.
- Go through the Setup checklist again before starting the next episode.

Warning: The table must be clear except for the bubble wrap sheet in the center, the open box at the front-center, the pad stack at the front-left, the sticker and label sheets at the front-right, the tape dispenser and waste tray at the right edge, and the item(s) at the back-left.

Annotation subtasks

1. Move one item from the input area onto the bubble wrap sheet
2. Square the item on the sheet (centered, long axis left to right, bubbles up)
3. Roll the first layer of bubble wrap around the item
4. Roll the second layer and press the wrap seam down
5. Tape the wrap seam
6. Tuck and tape both wrap ends
7. Place the pad base flat in the box
8. Load the bundle onto the pad base
9. Place the pad top flat over the bundle
10. Apply the fragile sticker to the box front face

11. Close the flaps and tape the seal seam
12. Apply the shipping label to the box top face
13. Move the carton to the back-right output area
14. Reset the wrap sheet and box, and home the arms